Conditional Generation of Lithium-Vacancy Configurations for Cathode Relithiation

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Abstract

We test whether conditional variational autoencoders (CVAEs) discover novel, low-energy lithium-vacancy configurations more efficiently than simple samplers. The study compares posterior encoders, causal spatial features, latent dimensions and a matched autoregressive model without a latent. On a fresh assessment, the selected CVAE, autoregressive control and mutation achieve 62.20%, 63.04% and 58.37% balanced useful yield under UMA. Spatial features help, but larger latent spaces improve reconstruction without consistent generation gains. Saved posterior statistics and controlled decoding probes reveal strong use of the explicit energy condition and weak incremental latent influence. A paired study of unrelaxed and relaxed energy labels does not improve relaxed-state sampling. These findings motivate controlled representation learning, rather than a claim of physical ground-state discovery.

1 Research question, motivation and related work

Direct cathode recycling restores active material instead of dissolving it into constituent elements; hydrothermal relithiation has regenerated spent $LiCoO_2$ (LCO) [1]. Li/vacancy arrangements differ in energy even at fixed composition [2]. Our longer-term motivation is to prioritize configurations for density functional theory (DFT) and recycling simulations. The *class question* is narrower: **does a CVAE improve novel low-energy sampling under an equal proposal budget, and which features drive the gain?** DFT supplies a specified electronic-structure reference; a machine-learned interatomic potential (MLIP) approximates energies and forces. A single-point evaluation keeps geometry fixed; relaxation moves atoms toward a local minimum.

Wakamiya and Ishikawa [3] combine a convolutional CVAE with Pt/Ni catalyst screening, conditioning on binary percentile labels for formation energy and oxygen-reduction overpotential. A padded four-layer occupation tensor and 32-dimensional latent model generate arrangements for iterative evaluation and retraining. Their evaluator already uses UMA, a universal MLIP [4]; replacing an "NNP" with an "MLIP" is not our novelty. We extend conditional atomistic generation to periodic, composition-constrained Li/vacancy patterns, explicitly testing posterior representation, spatial autoregression and latent utility. Symmetry-orbit novelty per *attempt*, no-latent controls and distinct relaxation targets make that question measurable. Our binary site representation needs no nonphysical padding.

2 Revised data and featurization

The proposal's 144-site pilot is replaced by an economical O3 parent, $Li_{48}Co_{48}O_{96}$. Learning masks $o \in \{0,1\}^{48}$ use *one for present Li*, reversing the proposal's vacancy-bit convention. Counts $n = \sum_i o_i \in \{12, 24, 36\}$ give $144 + n$ atoms and lithium fraction $x = n/48$. Deletion labels retain parent coordinates and cell. UMA-S-1.2.1/`omat` supplies training energies; MACE-MH-1/`omat_pbe` provides a second model check on the same proposals [4, 5]. Both use FP32, TF32 off.

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The parent's 192 retained permutations define symmetry *orbits*; the smallest transformed $1 - o$ string identifies each orbit. Grouping precedes splitting, and matched runs use canonical orientation. The original 40,157 TRAIN / 4,967 VAL / 5,123 TEST orbits retain their roles. Novelty uses a separately frozen history of 344,001 observed orbits. TEST and final assessment outcomes do not select models; equivalent views are not independent examples.

3 Conditional model and controlled alternatives

Energy labels and exact composition. Within each count and teacher, TRAIN energy quantiles q_{20}, q_{80} define LOW ($e \le q_{20}$), MIDDLE ($q_{20} < e \le q_{80}$) and HIGH, with energies in eV per present atom and ties unsplit. The four-dimensional condition is $c = (n/48, \mathrm{onehot}(b))$. A low request specifies a supported range, not a global minimum. Separate within-count thresholds avoid comparing raw energies across compositions.

The encoder defines $q_\phi(z \mid o, c) = \mathcal{N}(\mu_\phi, \mathrm{diag}(\sigma_\phi^2))$ and samples $z = \mu_\phi + \sigma_\phi \odot \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$. The generation prior $p(z) = \mathcal{N}(0, I)$ is fixed. A width-64 gated recurrent unit (GRU) decoder uses a fixed site order:

$$p_\theta(o \mid z, c) = \prod_{i=1}^{48} p_\theta(o_i \mid o_{<i}, z, c, h_i), \qquad \mathcal{L} = -\mathbb{E}_{q_\phi} \log p_\theta(o \mid z, c) + \beta_t D_{\mathrm{KL}}(q_\phi \| p). \quad (1)$$

Here h_i summarizes only the assigned prefix $o_{<i}$. Training uses true prefixes (teacher forcing); generation uses previous sampled bits. If no Li remain, later decisions are zero; if all remaining sites must be filled, they are one. Including this constraint in the likelihood and sampler enforces count without posthoc projection. Equation 1 follows conditional variational learning [6, 7]; both loss terms are in nats per mask.

What is featurized? The dense encoder uses 128→64 SiLU layers on the mask and condition. The graph encoder includes host atoms and vacant virtual Li sites, with element embeddings, mobile/presence flags and radial degree features. Two width-32 message-passing layers use six radial channels within 5 Å; separate host/mobile mean pools feed Gaussian heads [8]. Decoder context instead uses engineered causal fractions; it is not learned equivariant message passing.

Table 1: Matched model families and decoder featurization. All current CVAEs use $D = 4$.

Arm	Posterior	Decoder context beyond count/condition/site index
A	Graph Gaussian	Six occupied/vacant/unassigned fractions under two physical kernels.
B	None (AR)	Same six features; tests whether a latent is necessary.
C	Dense Gaussian	Same six features; tests a different posterior with the same decoder.
D	None (AR)	Same-size kernels under a fixed nonphysical site association.
A-shell/B-shell	Graph/none	Nine fractions: three categories in each of three radial shells.

Baseline kernels use nearest neighbors and exponential distance weights. Three shells at 2.811, 4.869 and 4.912 Å include interlayer neighbors and add 576 GRU parameters, so shell gains are not perfectly capacity-matched. D preserves B's parameter count and kernel degree/weight multisets but changes prefix exposure: it is not a pure locality ablation. A/C posterior capacities differ. Fixed ordering prevents full permutation equivariance; canonicalization handles retained symmetry.

Training and assessment protocol. Adam uses learning rate 10^{-3}, batch size 512 and 4,000 updates per full fit. TRAIN-only class weights reduce the 20/60/20 class imbalance. VAL retains its base weights and fixed Monte Carlo noise. The matched update-based warm-up is $\beta_t = 0.1 \min\{1, (\lfloor t/79 \rfloor + 1)/20\}$ for zero-based update t. Named checkpoints minimize the VAL reconstruction-plus-0.1KL objective. Thirty-seven new fits include controls, bounded screens, confirmations and six paired-target fits; they are not 37 replicates of one model.

Uniform sampling selects exactly n sites. Mutation makes one, two or three Li/vacancy swaps from a TRAIN seed in the requested class. Each method retains all attempted proposals, including repeats and failures. For replicate s, count n and class b, the primary yield is

$$Y_{snb} = \frac{\#\{\text{distinct valid orbits new to frozen history and measured in class } b\}}{\#\{\text{all attempts in cell } (s, n, b)\}}. \quad (2)$$

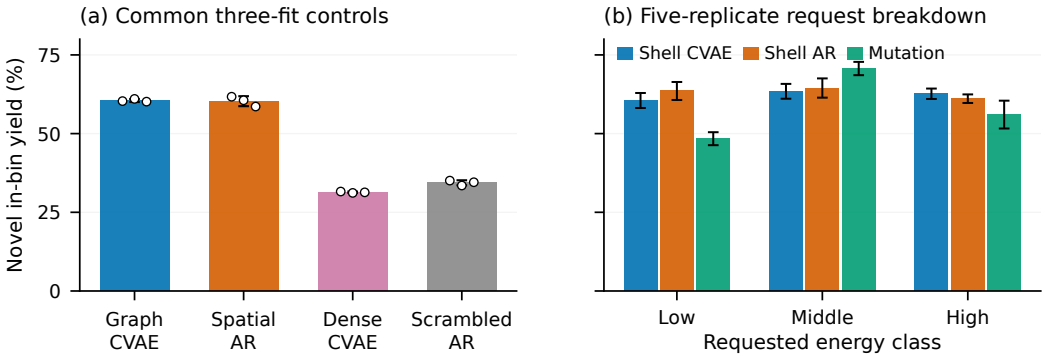


Figure 1: Fresh assessment of representation and requested-bin response. Left: common three-fit controls with identical decoder width. Right: selected shell models and mutation over five fit/sampling replicates. Bars show means, dots fits, and whiskers replicate SD (not confidence intervals). Mutation remains stronger on MIDDLE requests. All panels use UMA labels.

We average the nine count/class cells equally, then summarize independent fit/sampler replicates. Shared physical queries are scored once without changing attempt denominators. Each static count/bin/replicate cell gets 128 attempts; 44,928 primary attempts and 4,992 separate latent probes are frozen before scoring. The original fixed-q_{20} LOW score remains separate.

4 Featurization and hyperparameter sensitivity

Useful sampling does not require a variational posterior when physical decoder context is present (Figure 1). Common-three-fit graph CVAE minus spatial AR yield is $+0.20 \pm 1.49$ percentage points (pp); dense-posterior C and scrambled association D achieve only $31.37 \pm 0.22\%$ and $34.38 \pm 0.80\%$. On the complete five-fit comparison, adding shell features improves balanced UMA yield by 1.96 ± 0.32 pp for the CVAE and 2.83 ± 1.46 pp for AR. The selected CVAE does not outperform its matched no-latent control (difference -0.83 ± 1.13 pp). Standard deviations here describe paired fit differences; proposal rows are not independent fitted models.

Table 2: Fresh balanced novel in-request yield (%, mean $\pm$ replicate SD). All rows have five fit or sampling replicates. MACE evaluates the same UMA-trained proposals with its own frozen TRAIN thresholds.

Method	UMA	MACE
Graph CVAE, six features	60.24 ± 0.53	61.72 ± 0.72
AR, six features	60.21 ± 1.23	62.03 ± 0.76
Graph CVAE, shell features	62.20 ± 0.56	64.03 ± 0.79
AR, shell features	63.04 ± 1.47	64.38 ± 0.96
Mutation	58.37 ± 1.08	57.97 ± 1.39
Uniform fixed-count	33.32 ± 0.04	33.32 ± 0.04

What has and has not converged. The controlled $D \in \{4, 8, 16, 32\}$ sweep selected four as the economical latent size; it does not establish asymptotic convergence or optimality of $D = 4$ for every encoder. Reconstruction improves from 17.01 to 14.87 nats; yield remains 28.82–32.64%, near uniform and below the older mutation baseline (59.72%; Figure 2). A separate dense $D = 8$ development pool varies *training* class balance and *checkpoint-selection* KL weight:

TRAIN balance / selection β	Off/1	Off/0.1	On/1	On/0.1
UMA balanced yield (%)	29.28	30.44	30.61	32.12

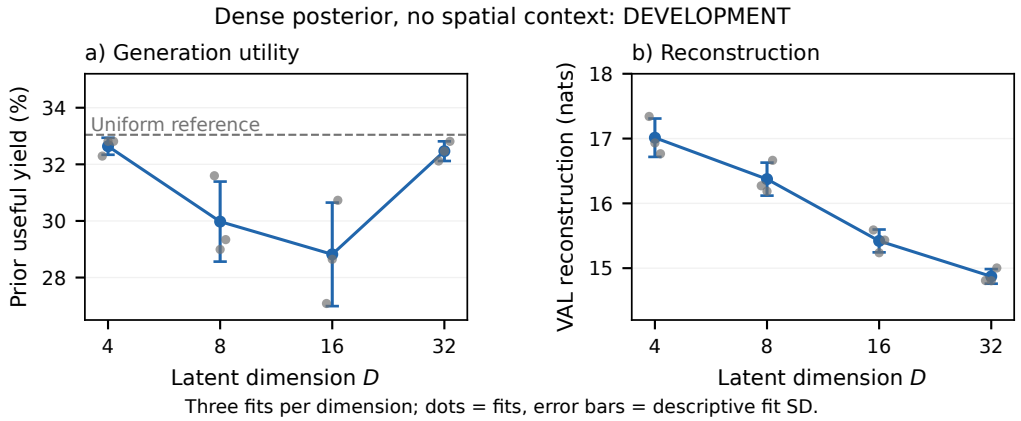


Figure 2: Earlier DEVELOPMENT sweep: dense posterior, no spatial decoder context, three fits per dimension. Larger D improves VAL reconstruction but does not monotonically improve prior-sampling yield. These data are not a dimension sweep of the newer graph/shell CVAE. Error bars are fit SD.

The training endpoint $\beta = 0.1$ is unchanged in that 2×2 comparison; selection β is not a training-β sweep. A newer single-fit $\beta = 0.03$ screen gains 0.69 pp balanced yield but loses 1.04 pp LOW yield and was not promoted. Remaining-count features do not beat the selected shell AR. Learning rate, batch size, widths, graph cutoff and the 4,000-update budget are fixed settings, not demonstrated convergence results. Cyclical KL annealing has not been executed. Development outcomes do not add held-out replicates.

5 What the trained latent space actually represents

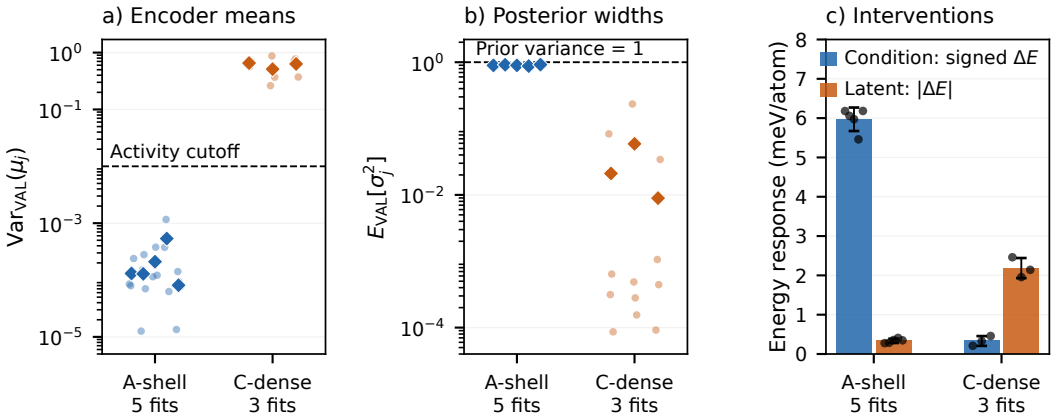


Figure 3: Latent diagnostics: selected A-shell (five fits) versus C with legacy context (three fits), not a pure posterior ablation. Component points are not independent fits. The 0.01 activity cutoff is heuristic. Interventions fix decoder uniforms: LOW→HIGH signed shifts and alternate-z absolute shifts are different observables. Diamonds and energy error bars denote fit means and SD.

A broad posterior can encode very little. The selected shell CVAE has VAL KL 0.0072–0.0202 nats/mask across five fits (mean 0.0123). Posterior-mean variances over masks span 1.26×10^{-5}–1.16×10^{-3}; mean posterior variances $\mathbb{E}[\sigma_j^2]$ are 0.854–0.943. All fits have zero active coordinates

at the heuristic $\mathrm{Var}(\mu_j) > 0.01$ cutoff; only one coordinate in one fit exceeds 0.001. These broad, almost input-independent posteriors exhibit *posterior collapse*, not point-like or zero-entropy codes.

A conditional information decomposition makes the interpretation more precise [9]. Under the same weighted empirical VAL distribution used for the saved KL summaries, define $q(z \mid c) = \mathbb{E}_{o|c} q_\phi(z \mid o, c)$. Then

$$\overline{K} \equiv \mathbb{E}_{o,c} D_{\mathrm{KL}}(q_\phi(z \mid o, c) \| p(z)) = I_q(o; z \mid c) + \mathbb{E}_c D_{\mathrm{KL}}(q(z \mid c) \| p(z)). \tag{3}$$

Adding and subtracting $\log q(z \mid c)$ inside the KL gives this identity; nonnegativity gives $I_q(o; z \mid c) \leq \overline{K}$. The largest observed shell KL therefore bounds this empirical conditional information by approximately 0.030 bits/mask ($\overline{K} / \log 2$). This is an upper bound, not an MI estimate, a population guarantee or a physical configurational entropy. Large KL supplies no corresponding lower bound: it can also reflect aggregate-posterior mismatch.

Encoding, conditioning and sampling are different channels. Generation can be written $o = F_\theta(c, z, U)$, where U contains independent uniform numbers for categorical AR draws. Holding the actual z, U fixed while changing LOW to HIGH shifts UMA energy by 5.97 meV/atom for shell A. Resampling only z at fixed c, U changes mean absolute energy by just 0.335 meV/atom and 0.851 occupation bits; 65.42% of paired outputs remain in the same orbit. Thus the explicit condition has a clear effect, while the latent's incremental influence is weak. A no-latent AR remains stochastic through U, explaining why weak latent use does not imply identical samples.

C has 12.52–13.32 nats KL and four active coordinates, yet poor prior-generation yield. It changes 13.04 bits when resampling z, but its LOW→HIGH response is only 0.331 meV/atom. An informative-looking encoder or sharp posterior is not sufficient for useful *conditional prior sampling*. Signed mean energy change from two exchangeable prior draws should be near zero; absolute/RMS changes, orbit identity and condition response are more discriminating. Latent coordinates have no assigned physical meaning and can change sign or order across fits.

6 Relaxation target, limitations and next steps

A matched target test. A separate 6,000-mask cohort has 3,600 TRAIN / 1,200 VAL / 1,200 TEST roles assigned before labels. For each mask, E_{del} uses parent coordinates and E_{end} follows fixed-cell UMA relaxation ($F_{\max} < 0.05$ eV/Å; 500-step/120-second caps). Eight interruptions remain recorded; fits use 3,594 usable TRAIN and 1,200 VAL rows. Three CVAEs per target share graph/shell architecture, initialization and 4,000 updates, but use separate TRAIN-fitted thresholds. Endpoint conditioning still generates a mask: coordinates require relaxation.

The frozen comparison retains 10,944 attempts and 7,616 unique queries, all converged. Original six-feature A/B/mutation endpoint LOW yields are 65.63/64.06/53.82% (three replicates); CVAE minus AR is 1.56 ± 2.76 pp. In the matched new-cohort comparison, endpoint-label training gives 60.42% endpoint LOW yield versus 62.50% for deletion-label training (-2.08 ± 2.76 pp; no positive replicate). All-bin endpoint yields are 58.74% versus 59.14%. Thus initial-energy control survives relaxation, but endpoint-label training does not improve this bounded comparison. Original and new-cohort arms use different data and contexts; their difference is not an architecture effect. Applying deletion thresholds to shifted endpoint energies would falsely inflate success.

Advantages and limits. Exact counts, symmetry-orbit novelty, all-attempt denominators and matched no-latent controls make representation gains interpretable. Evidence nevertheless covers one parent, three compositions and two teachers, not new-size/material transfer or DFT accuracy. Low-lithiation O3 structures are a configuration benchmark, not stable-phase predictions. The model is not fully equivariant, not every ablation is capacity-matched, and few fits do not establish equivalence or global hyperparameter convergence. Low-quantile requests are not ground-state guarantees; generated frequencies are not equilibrium probabilities. Full-TRAIN orbit checks establish nonidentity, not distinct physical minima.

Remaining class work. Beihao will reproduce training/latent diagnostics; Varad will audit metrics and fit variability; Ricardo will review structures and physical interpretation. Further latent interventions use development data and a newly frozen assessment, preserving TEST roles. Joint-coordinate generation, DFT/MLIP fine-tuning and size transfer remain extensions.

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